

Chapter 5 – The MOSFET

Metal Oxide Semiconductor Field Effect Transistor

5.1 MOS Structure

5.2 MOSFET Basics & Device Physics

5.3 MOSFET Circuit Models

5.4 SPICE Analysis

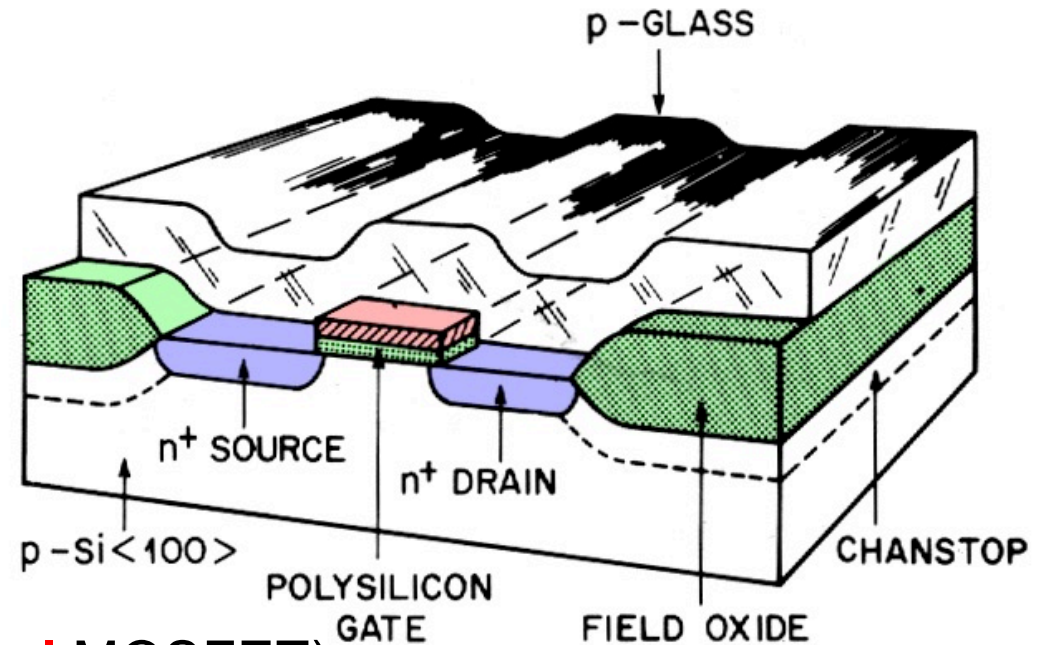
Literature: Pierret, Chapter 16&17, page 563-637

Jaeger, Blalock, Chapter 4, page 145-202

MOSFET Device Structure

Metal Oxide Semiconductor Field Effect Transistor

- Most important device structure in VLSI
- First MOSFET: 1960
- 4-terminal device: gate, substrate, drain, source



Basic Device Principle (n-channel MOSFET)

- Positive gate voltage **depletes** holes underneath the gate oxide
- $V_g > V_T$ (threshold voltage): large electron concentration (**inversion**) underneath gate oxide: $n_p > p_{p0}$
- As a result, a **conducting channel** is formed between source and drain, resulting in a current flow in case $V_D \neq 0$

➡ **Switching Device:** ON: $V_g > V_T$; OFF: $V_g < V_T$

5.1 MOS Capacitor

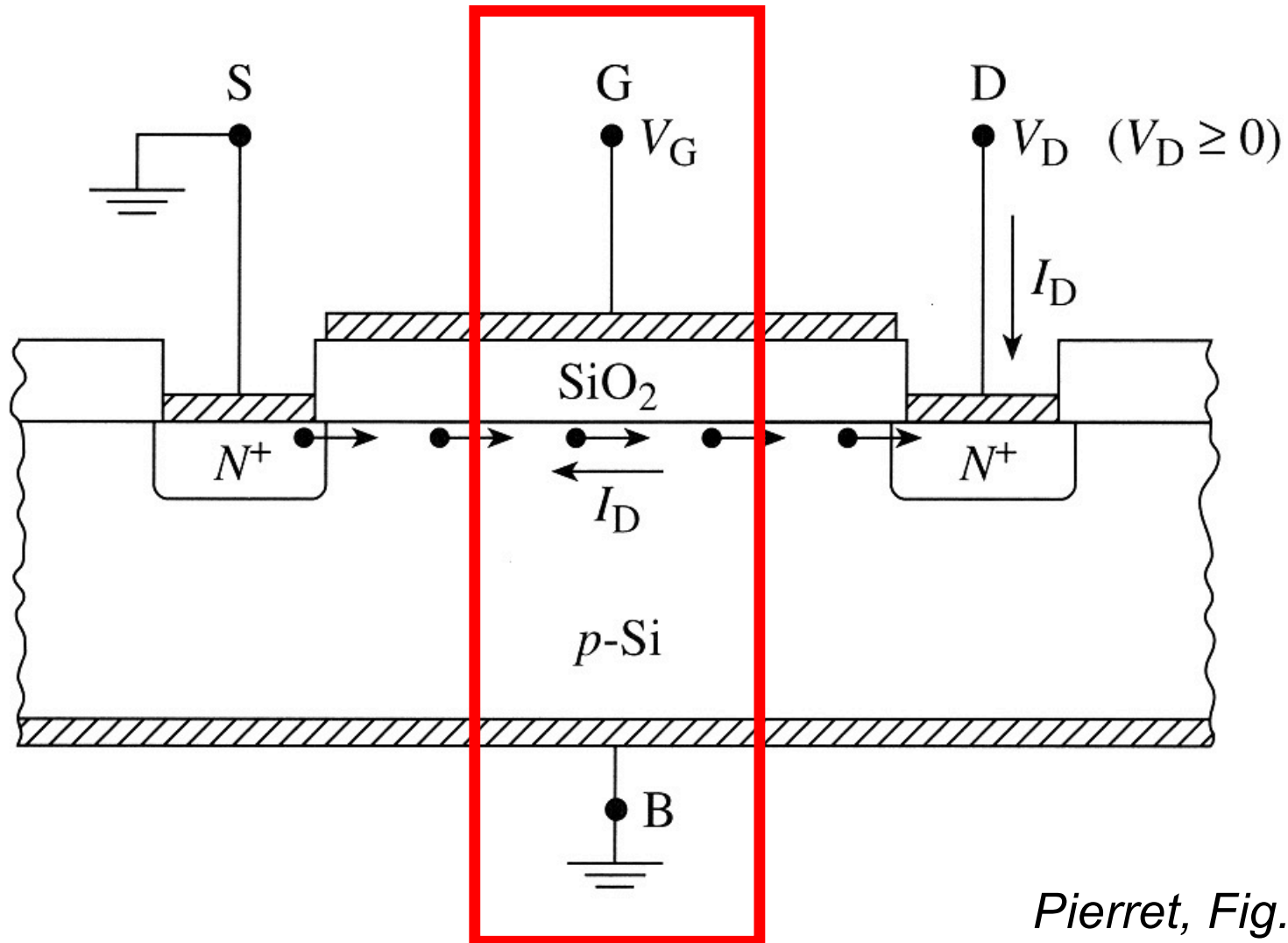
5.1.1 Band Diagram

5.1.2 Electrostatics

5.1.3 The Si-SiO₂ MOS Capacitor

Literature: Pierret, Chapter 16, page 563-599

MOS Capacitor



Pierret, Fig. 17.1

5.1.1 Energy Band Diagram

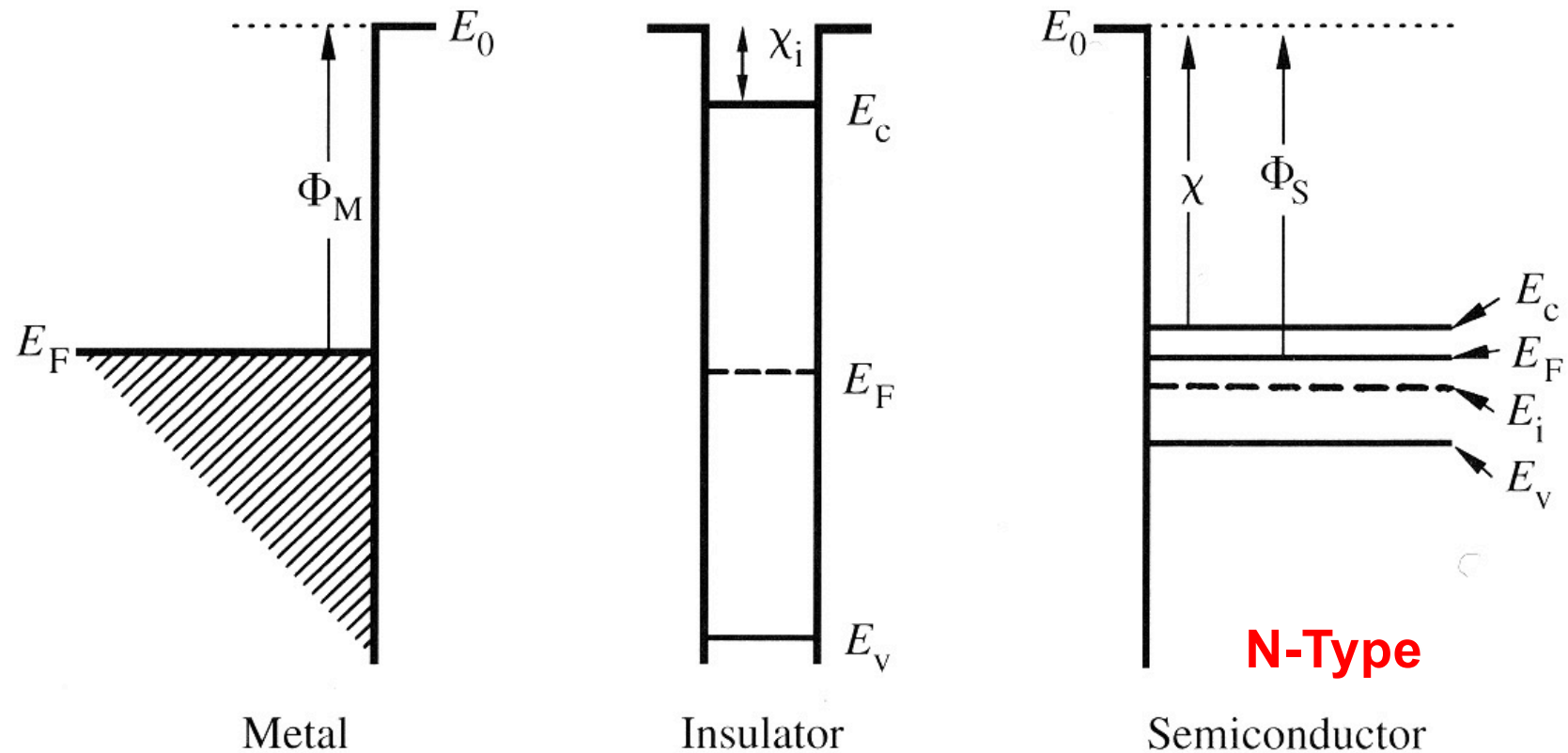
Assumptions:

- Thick metallic gate, i.e. equipotential region
- Perfect insulator, i.e. no dc current
- No charges in oxide or interface
- Uniformly doped semiconductor
- Sufficiently thick semiconductor having field-free region close to contact
- One-dimensional structure
- **Flat-band condition:** $\Phi_M = \Phi_S = \chi + (E_C - E_F)_{FB}$

Work function Φ : Energy difference between vacuum level E_0 and Fermi Energy E_F

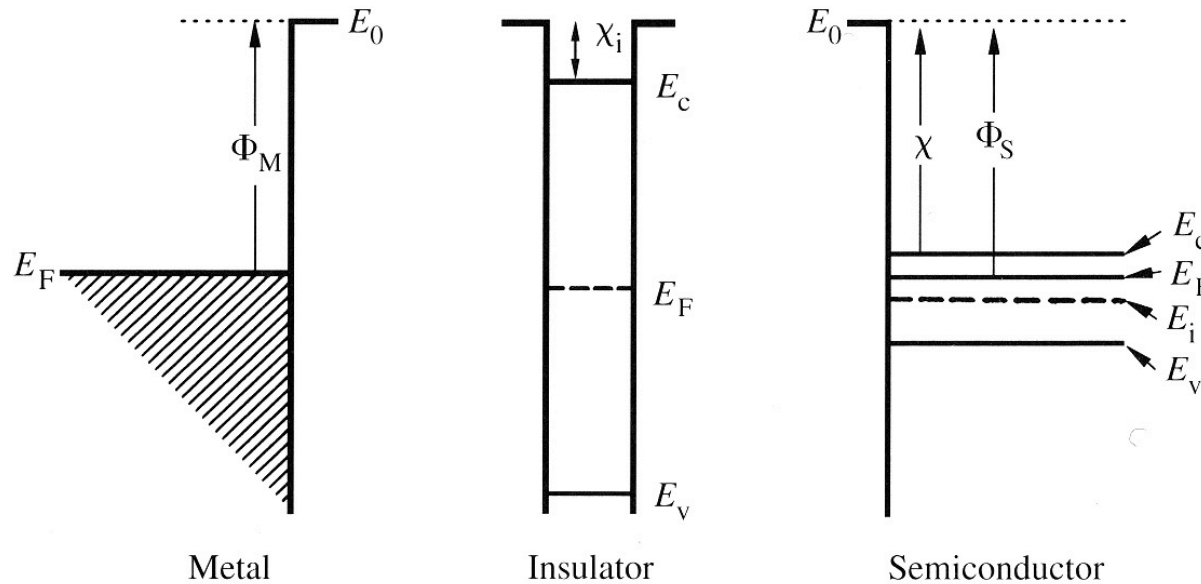
Electron affinity χ : Energy difference between vacuum level E_0 and conduction band energy E_C

Band Structure of Individual Materials



Pierret, Fig. 16.2

Work Function & Electron Affinity



Pierret, Fig. 16.2

E_0 vacuum level

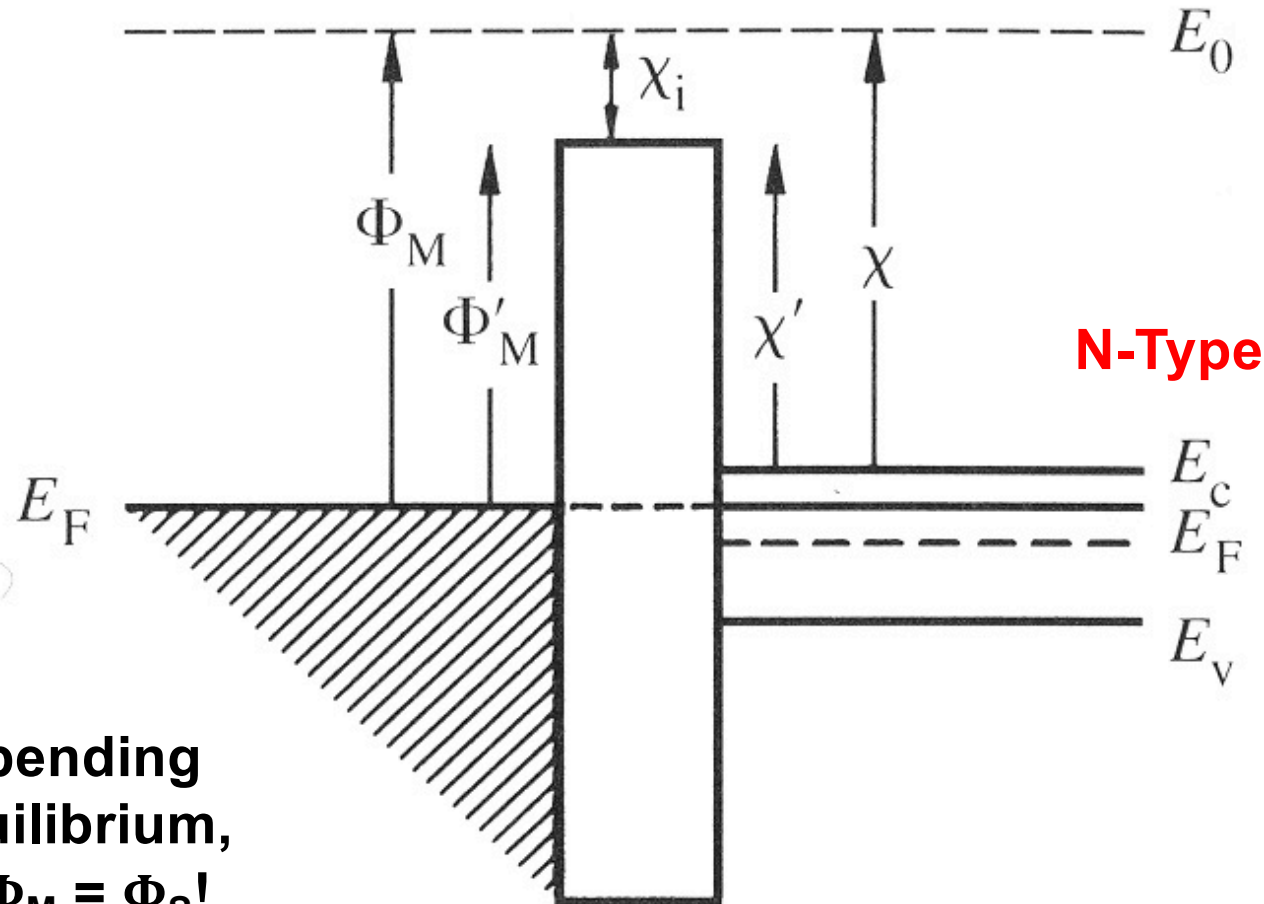
Φ_M metal work function (fundamental property of particular metal)
 = 4.28 eV for Al = 4.33 eV for Ti = 4.50 eV for Cr
 = 4.55 eV for W = 5.10 eV for Au = 5.65 eV for Pt

Φ_S semiconductor work function (depends on material & doping)

χ electron affinity = $E_0 - E_C$ (fundamental property of semiconductor)
 = 4.03 eV for Si = 4.07 eV for GaAs = 4.00 eV for Ge

MOS Capacitor

– Equilibrium Energy Band Diagram –



No band bending
under equilibrium,
because $\Phi_M = \Phi_S$!

Pierret, Fig. 16.3

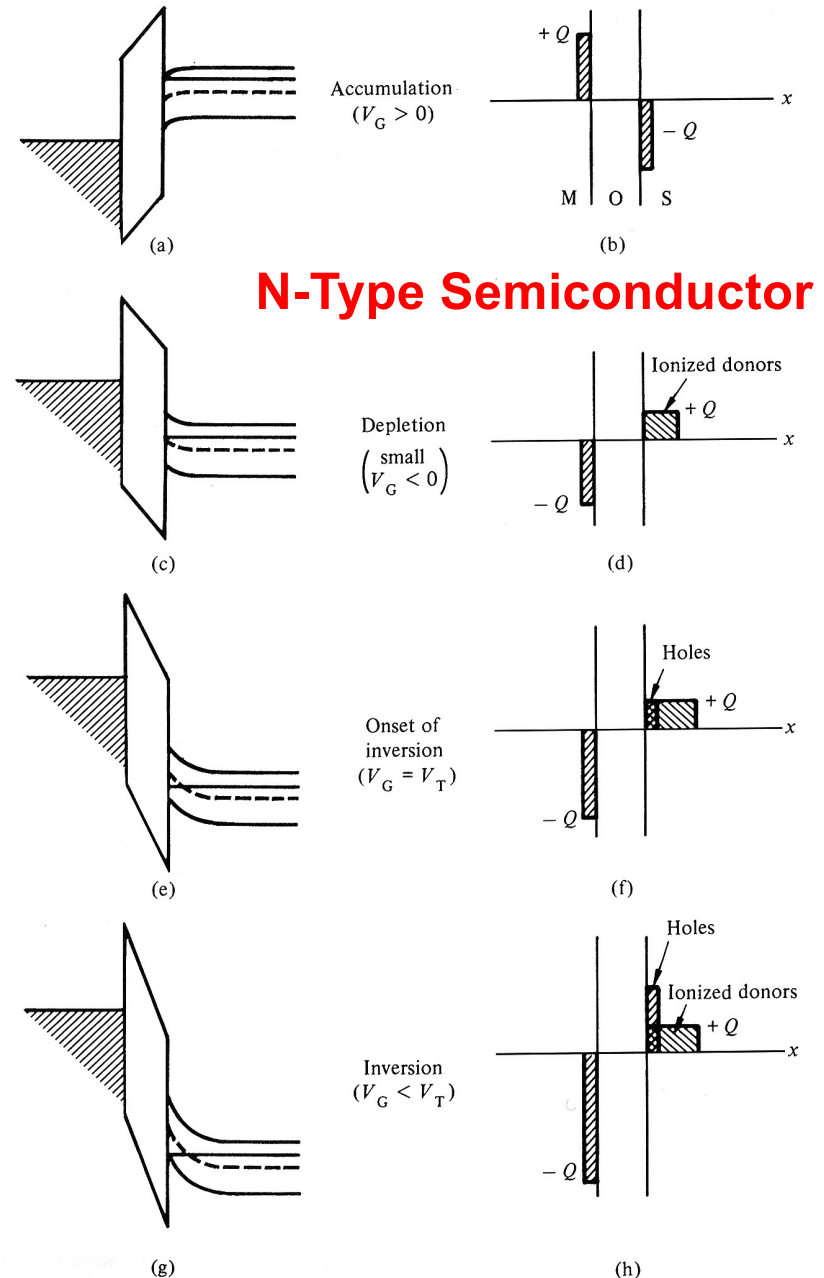
MOS Capacitor Biased Band Diagram

- Bias V_g applied to gate electrode
- Fermi energy:

$$E_F(\text{metal}) - E_F(\text{semicon.}) = -qV_g$$

Note:

- **E_F is constant inside the semiconductor** because there is no dc carrier flow through the insulator
- **No band bending inside the metal** because it is considered an equipotential region
- **Constant energy slope inside insulator** because there are no charges inside it and thus the electric field is constant across it



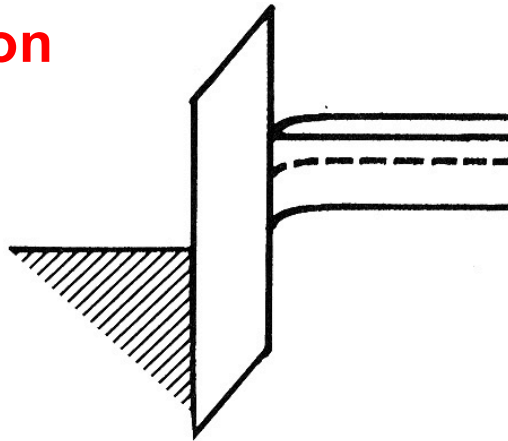
N-Type Semiconductor

Pierret, Fig. 16.5

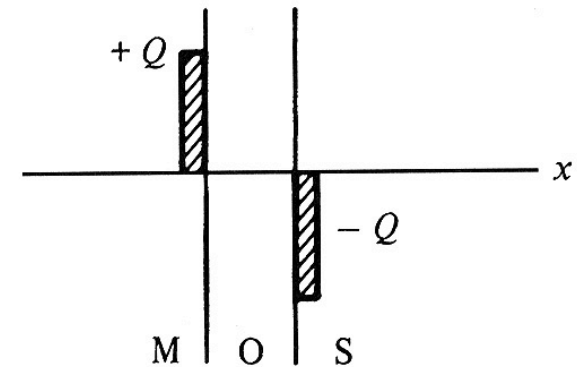
MOS Capacitor (N-Type)

– Biasing Regions I –

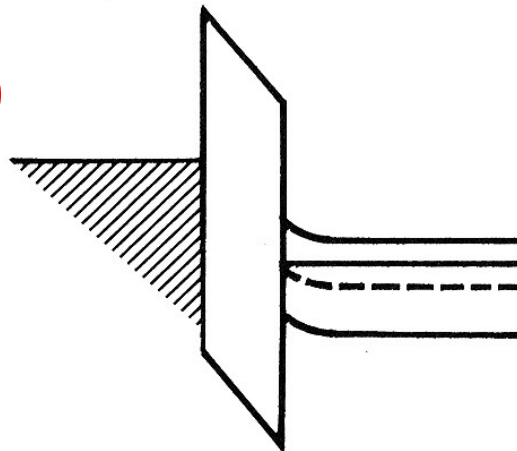
Accumulation
 $V_g > 0$



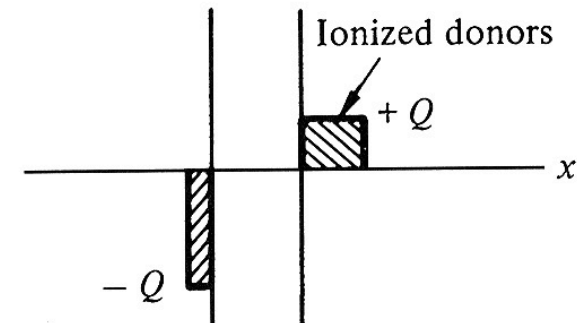
Accumulation
($V_G > 0$)



Depletion
small $V_g < 0$



Depletion
(small
 $V_G < 0$)



Pierret, Fig. 16.5

MOS Capacitor (N-Type)

– Biasing Regions II –

Onset of Inversion

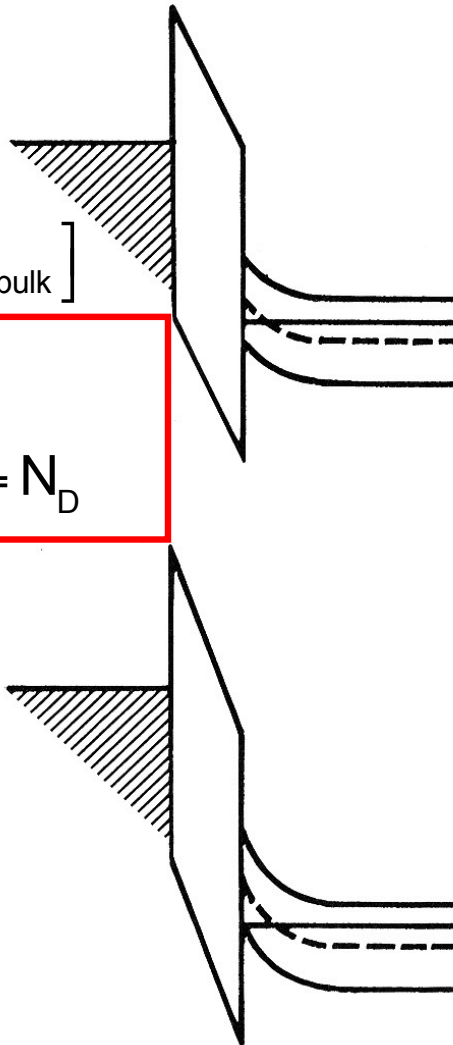
$$V_g = V_T$$

$$E_{i,\text{surface}} - E_{i,\text{bulk}} = 2[E_F - E_{i,\text{bulk}}]$$

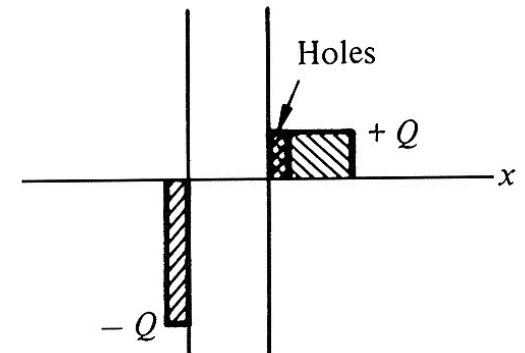
$$\begin{aligned} p_s &= n_i e^{(E_{i,\text{surface}} - E_F)/kT} \\ &= n_i e^{(E_F - E_{i,\text{bulk}})/kT} = n_{\text{bulk}} = N_D \end{aligned}$$

Inversion

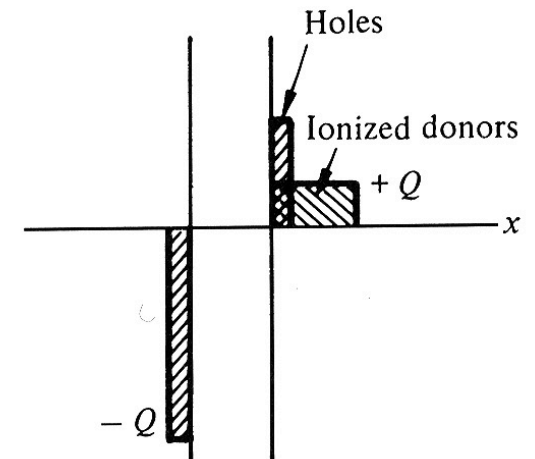
$$V_g < V_T$$



Onset of
inversion
($V_G = V_T$)



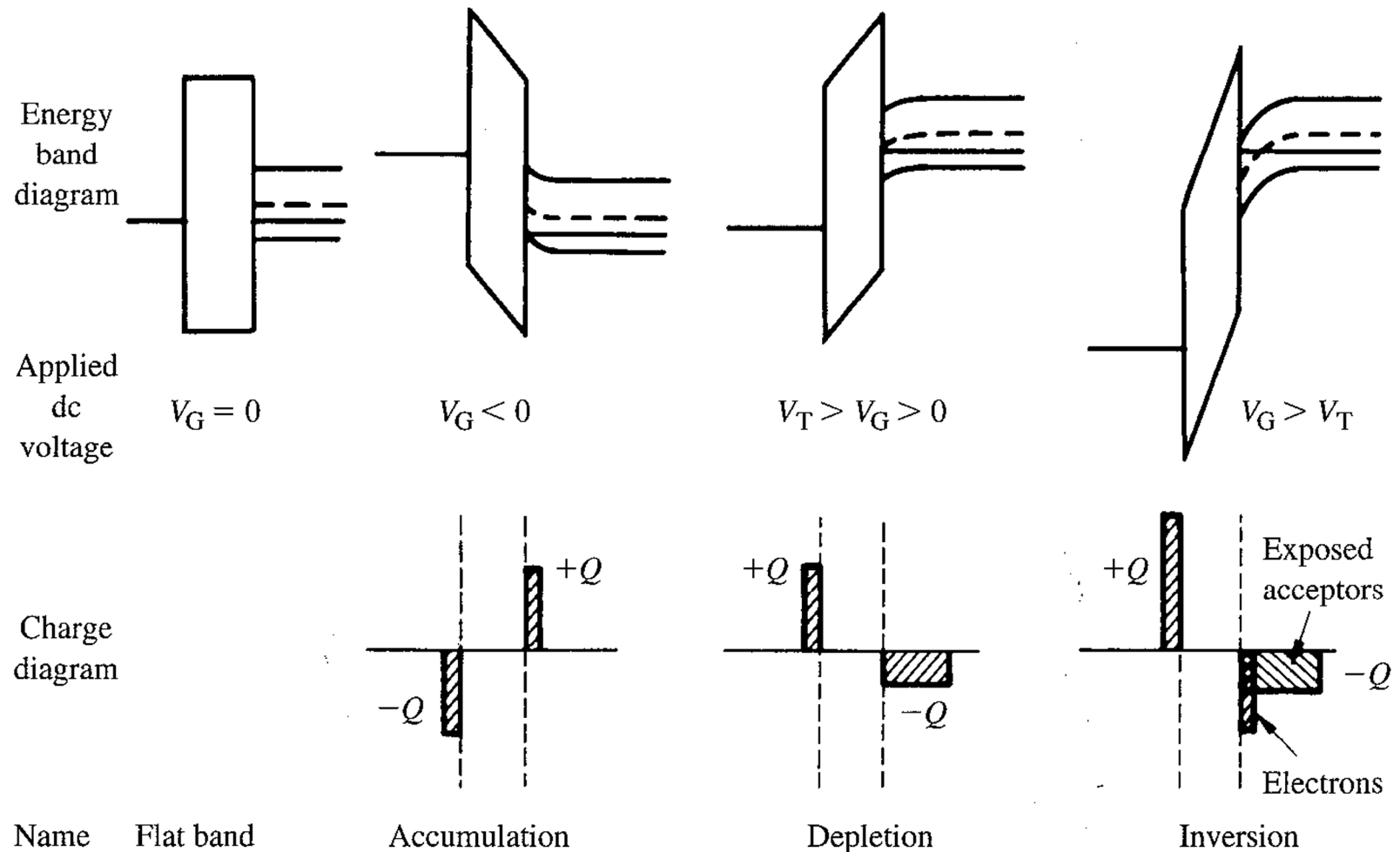
Inversion
($V_G < V_T$)



Pierret, Fig. 16.5

MOS Capacitor (P-Type)

– Biasing Regions –



Pierret, Fig. 16.6

5.1.2 MOS Capacitor – Electrostatics

- Potential $\phi(x)$ inside semiconductor

$$\phi(x) = -\frac{1}{q} [E_i(x) - E_{i,bulk}]$$

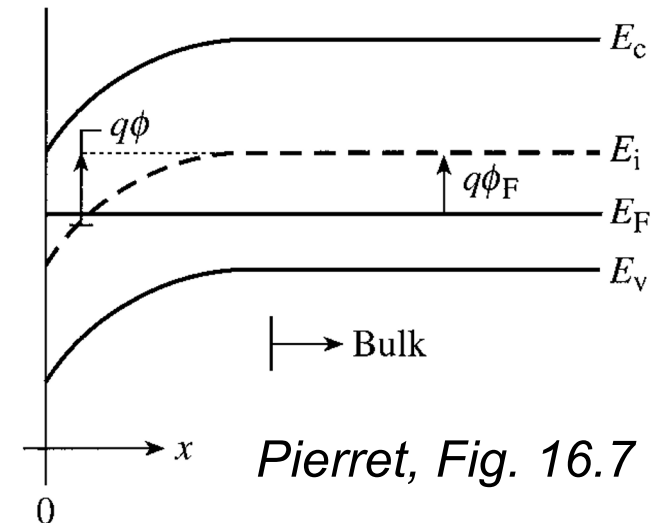
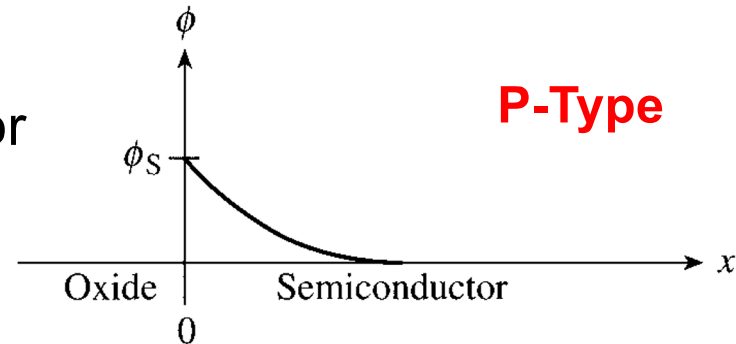
- Surface potential ϕ_s

$$\phi_s = \frac{1}{q} [E_{i,bulk} - E_{i,surface}]$$

- Fermi potential ϕ_F

$$\phi_F \equiv \frac{1}{q} [E_{i,bulk} - E_F] \begin{cases} > 0 \text{ for p type} \\ < 0 \text{ for n type} \end{cases}$$

$$\phi_F = \begin{cases} +\frac{kT}{q} \ln \left[\frac{N_A}{n_i} \right] & \text{for p type} \\ -\frac{kT}{q} \ln \left[\frac{N_D}{n_i} \right] & \text{for n type} \end{cases}$$



$\phi_s = 2 \phi_F$ at onset of inversion
 $\phi_s > 2 \phi_F$ in inversion
 $0 < \phi_s < 2 \phi_F$ in depletion

MOS Capacitor Electrostatics

– Depletion Region (P-Type Semiconductor) –

- **Assumptions:**
 - **P-type semiconductor**
 - **Constant charge density** in depletion region:
 $\rho = -q N_A$ for $0 \leq x \leq W$
- **Electric Field** (boundary condition: $\mathcal{E}(W) = 0$):

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{K_s \epsilon_0} \approx -\frac{qN_A}{K_s \epsilon_0} \Rightarrow \boxed{\mathcal{E}(x) = \frac{qN_A}{K_s \epsilon_0} (W - x)}$$

- **Electrostatic Potential** (boundary condition: $\phi(W) = 0$):

$$\mathcal{E} = -\nabla\phi \Rightarrow \boxed{\phi(x) = \frac{qN_A}{2K_s \epsilon_0} (W - x)^2}$$

MOS Capacitor Electrostatics

– Depletion Region (cont.) –

- **Surface Potential $\phi_s = \phi(0)$:**

$$\phi_s = \frac{qN_A}{2K_s\epsilon_0} W^2$$

- **Width of Depletion Region W and Maximum Width at Onset of Inversion W_T :**

$$W = \sqrt{\frac{2K_s\epsilon_0}{qN_A} \phi_s}$$
$$W_T = \sqrt{\frac{2K_s\epsilon_0}{qN_A} 2\phi_F} = \sqrt{\frac{4K_s\epsilon_0 kT}{q^2 N_A} \ln\left[\frac{N_A}{n_i}\right]}$$

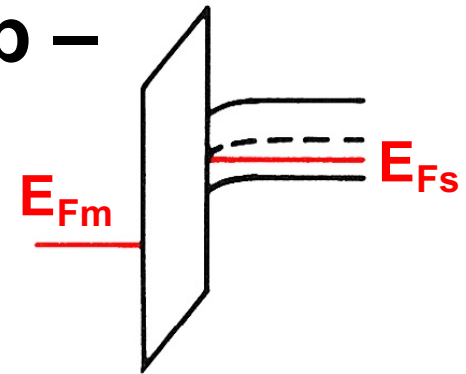
- How does ϕ_s depend on V_g in depletion mode?

MOS Capacitor Electrostatics

– Gate Voltage Relationship –

- How does ϕ_s depend on V_g in depletion mode?

$$V_g = \underbrace{\Delta\phi_{\text{semi}}}_{=\phi_s} + \Delta\phi_{\text{oxide}} = \phi_s + \Delta\phi_{\text{oxide}}$$



- No charges inside the oxide with thickness x_0 , i.e. $\rho_{\text{oxide}} = 0$, yields

$$\frac{d\epsilon_{\text{oxide}}}{dx} = \frac{\rho_{\text{oxide}}}{K_o \epsilon_0} = 0 \Rightarrow \epsilon_{\text{oxide}} = -\nabla\phi_{\text{oxide}} = \text{const.}$$

$$\Delta\phi_{\text{oxide}} = \int_{-x_0}^0 \epsilon_{\text{oxide}} dx = x_0 \epsilon_{\text{oxide}}$$

- The electric field inside the oxide can be obtained from the boundary condition stating that the **dielectric displacement** D at the oxide-semiconductor interface must be continuous:

$$D_{\text{oxide}} = D_{\text{semi}, x=0} \Rightarrow \epsilon_{\text{oxide}} = \frac{K_s}{K_o} \epsilon_{\text{semi}}$$

$K_s = 11.8$ for Si $K_o = 3.9$ for SiO_2

MOS Capacitor Electrostatics

– Gate Voltage Relationship (cont.) –

- Resulting in a **gate voltage V_g** in the depletion region

$$V_g = \phi_s + \frac{K_s}{K_o} x_0 \epsilon_{\text{semi}}(0)$$

$$V_g = \phi_s + \frac{K_s}{K_o} x_0 \sqrt{\frac{2qN_A}{K_s \epsilon_0} \phi_s} \quad 0 \leq \phi_s \leq 2 \phi_F$$

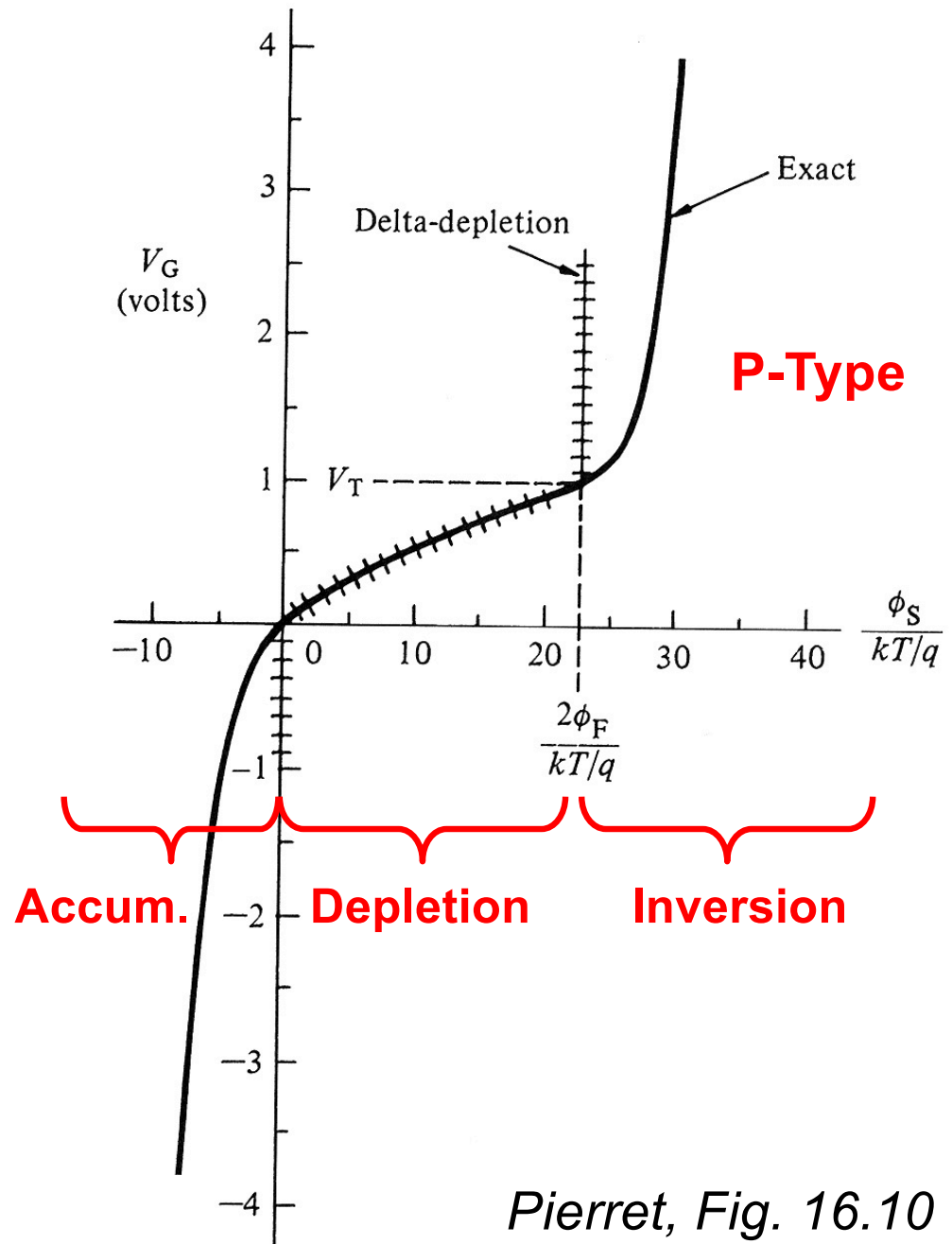
with the electric field at the semiconductor surface

$$\begin{aligned} \epsilon_{\text{semi}}(0) &= \frac{qN_A}{K_s \epsilon_0} W = \frac{qN_A}{K_s \epsilon_0} \sqrt{\frac{2K_s \epsilon_0}{qN_A} \phi_s} \\ &= \sqrt{\frac{2qN_A}{K_s \epsilon_0} \phi_s} \end{aligned}$$

MOS Capacitor

Gate Voltage V_g vs. Surface Potential ϕ_s

- Surface potential is rather rapidly varying with V_g in the depletion region, but not in the accumulation or inversion region
- In the accumulation and inversion regions, changes in the applied voltage are dropped almost totally across the oxide



Pierret, Fig. 16.10

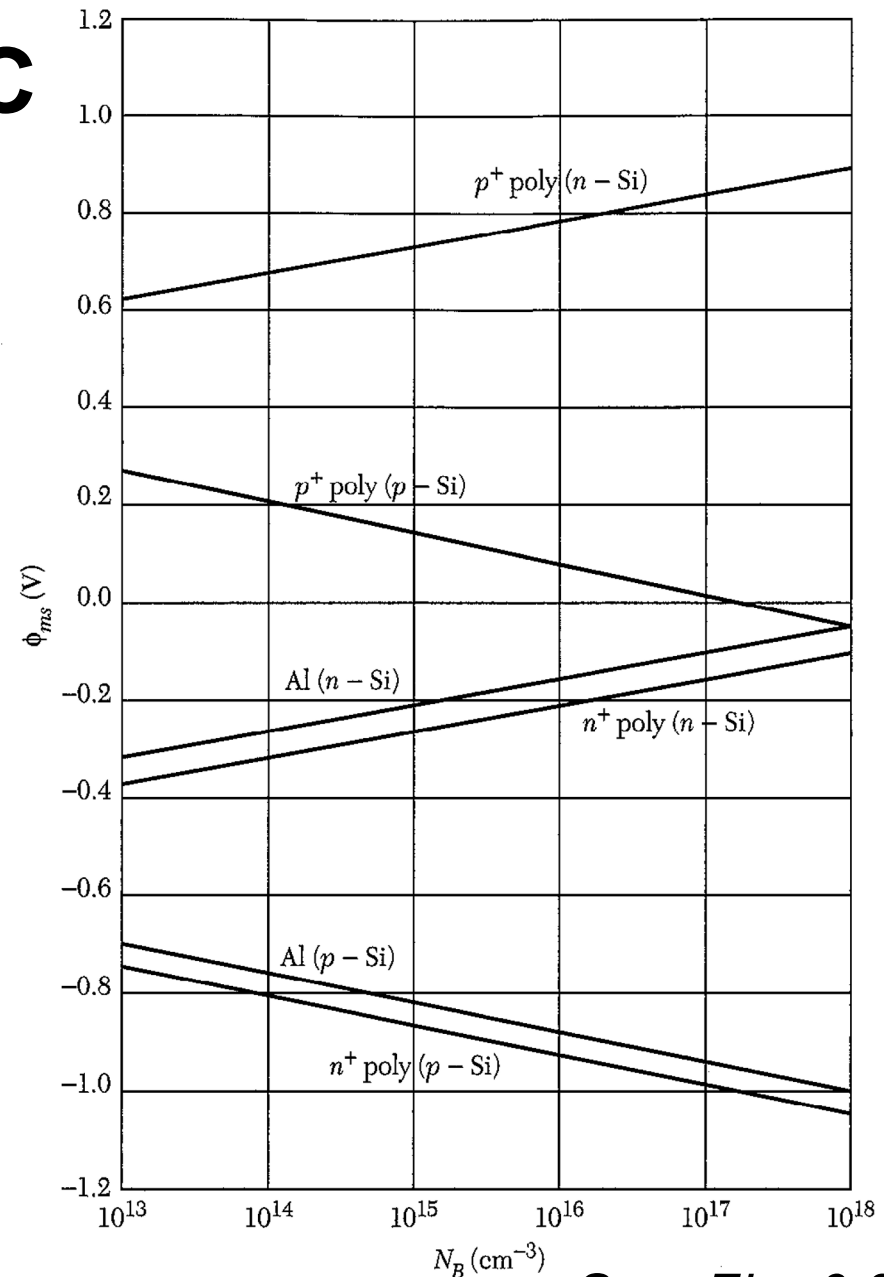
5.1.3 Si-SiO₂ MOS-C

– Work Function Difference –

- Metal-SiO₂-Si is the most extensively studied MOS capacitor
- For commonly used metal electrodes (e.g. polysilicon and aluminum), the work function difference is not zero:

$$\Phi_{MS} = q\phi_{MS} = \Phi_M - \Phi_S \neq 0$$

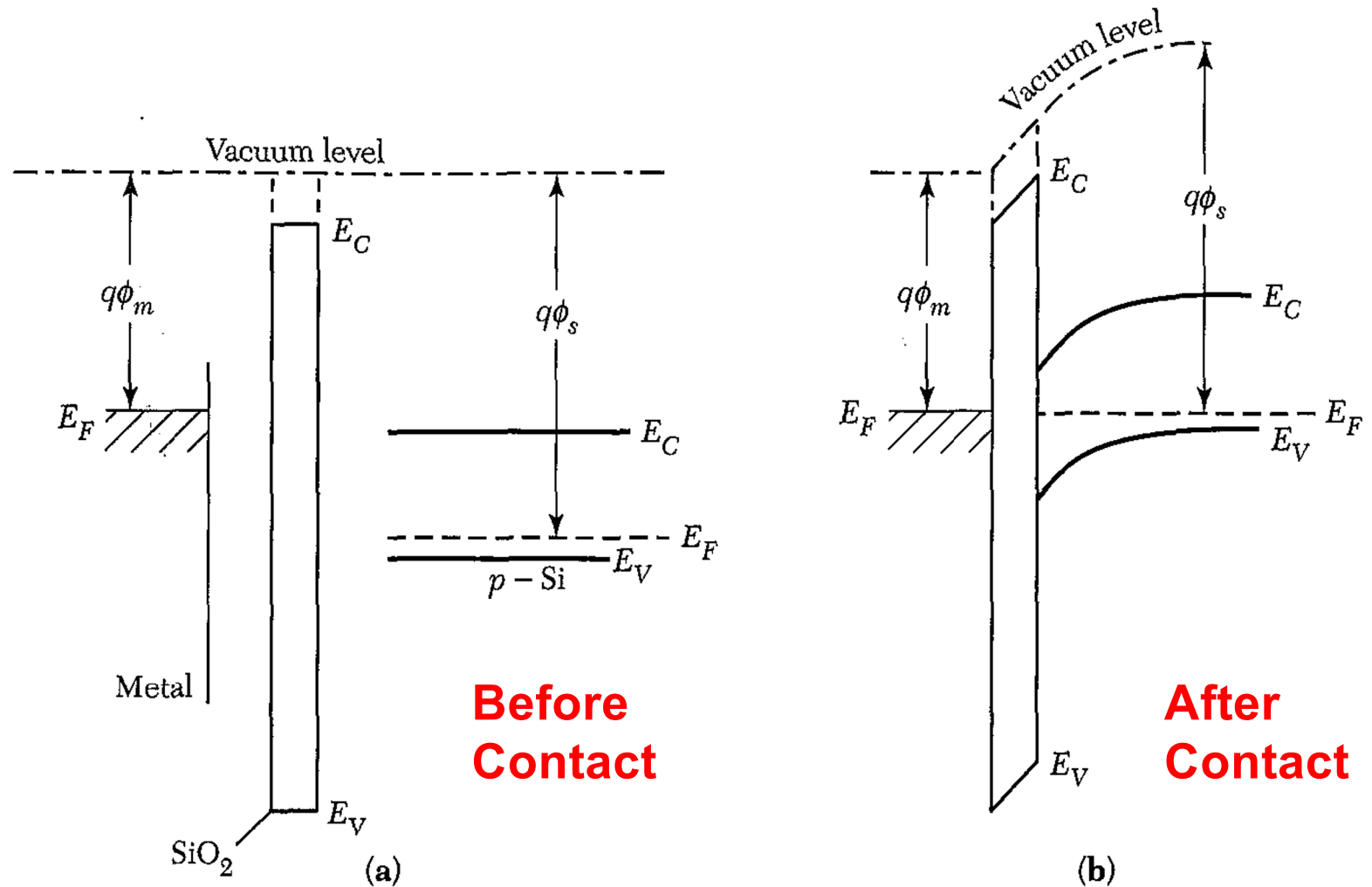
- The work function difference results in a band bending in the Si and SiO₂ even in equilibrium, i.e. $V_g = 0$
- In order to reach flat-band conditions, the so called **flat-band voltage** V_{FB} has to be applied to the gate: $V_{FB} = \Phi_{MS}$



Sze, Fig. 6.8

Si-SiO₂ MOS Capacitor

– Work Function Difference –

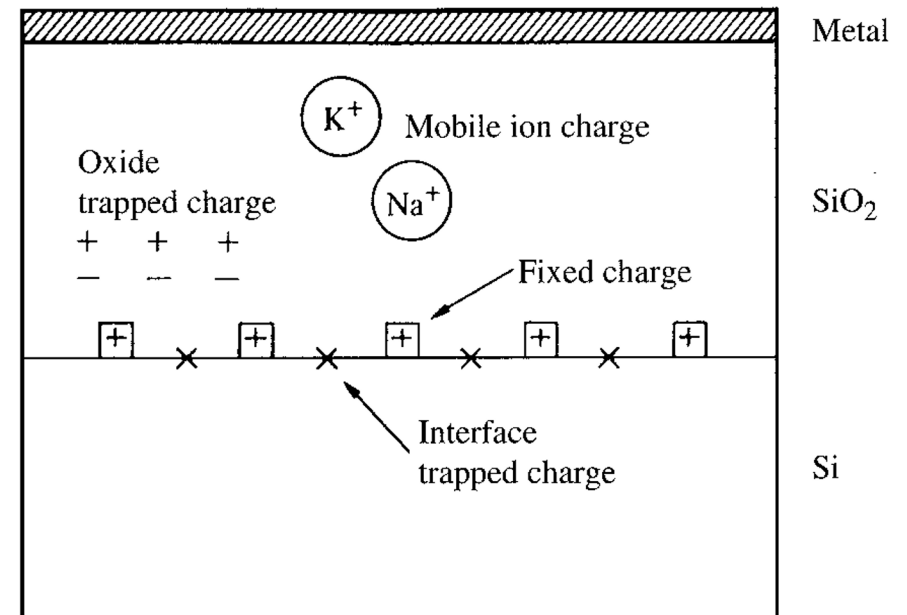


Sze, Fig. 6.9

Si-SiO₂ MOS Capacitor

– Interface Traps and Oxide Charges –

- In addition to the work function difference, the MOS capacitor is affected by charges in the oxide and traps at the Si-SiO₂ interface
- Types of interface traps and oxide charges:
 - Interface-trapped charges Q_{it}
 - Fixed oxide charges Q_f
 - Oxide-trapped charges Q_{ot}
 - Mobile ionic charges Q_m



Pierret, Fig. 18.4

Si-SiO₂ MOS Capacitor

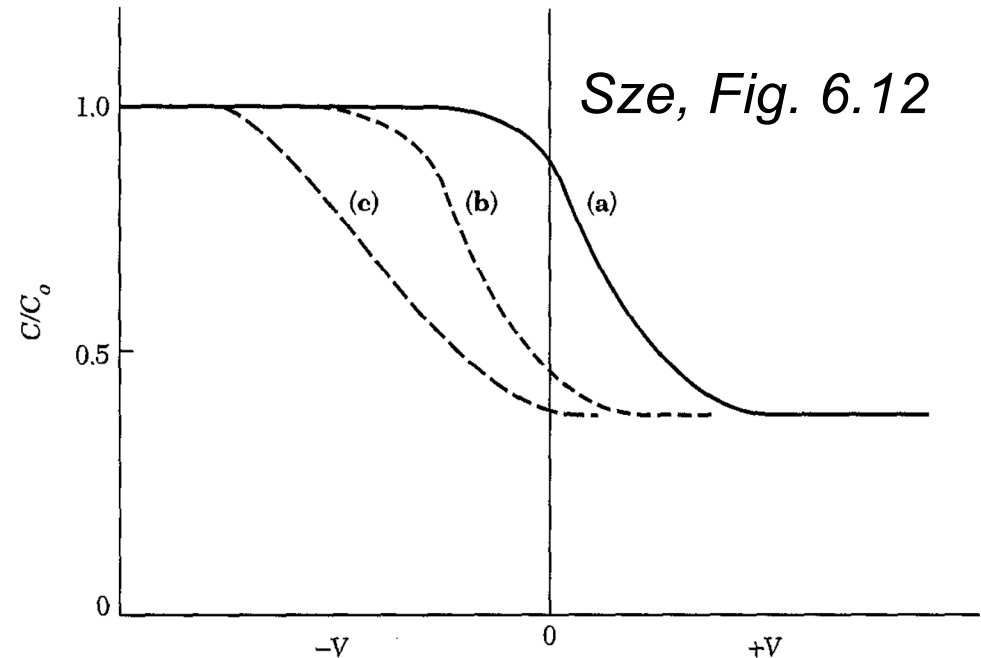
– Oxide Charges and Flat-Band Voltage –

- Influence of charges on the flat-band voltage depends on location of charges within the oxide:
 - Minimal influence at the metal-SiO₂ interface
 - Maximal influence at the Si-SiO₂ interface

$$V_{FB} = -\frac{1}{C_o} \left[\frac{1}{x_0} \int_0^{x_0} x \rho(x) dx \right]$$

- Often, the flat band voltage is expressed by the effective oxide charges

$$V_{FB} = \phi_{MS} - \frac{Q_f + Q_m + Q_{ot}}{C_o} \quad \text{with e.g.} \quad Q_m = \frac{1}{x_0} \int_0^{x_0} x \rho_m(x) dx$$



(a) Ideal MOS-C

(b) Parallel shift due to oxide charges

(c) Slope change due to interface-trapped charges